A Comparative Study of Energy Consumption for Large Language Models

William Yu
Brown University
william_yu@brown.edu

May 14, 2026

Abstract

Within the recent proliferation of Large Language Models lies the early innings of agentic artificial intelligence. Through the voracious consumer and enterprise appetite for tools like Claude Code, Cursor, and Copilot, coding has become the proving ground for agentic work. In doing so, it reveals critical implications for the future of inference, tokens, energy, and broadly speaking, human interaction with AI. From individual coders to companies, AI coding tools have been refined for orchestration, ever-increasing context lengths, and complex chain-of-thought.

Significant attention and concern has gone into the financial implications of AI. However, less focus has been paid to the physical energy costs of this new paradigm. We investigate the statistical profiles of energy consumption for Large Language Models, focusing on tail risks in agentic workflows.

Varying in model architecture and hardware, we fit distributions and assess differences in power draw. In doing so, we provide a generalized process for analyzing practical considerations and trade-offs in model and hardware selection. Our results indicate that newer GPUs do improve throughput and energy efficiency. Although a Gamma Likelihood Ratio Test on individual runs detects a statistically significant difference in distributional shape between reasoning and non-reasoning models ($\chi^2 = 22.697$, $p < 0.0001$), a group-mean comparison finds no significant difference in per-token energy intensity, with the non-result attributable to the small reasoning group size ($n = 4$) and limited statistical power. Using a Gamma Generalized Linear Model, we conclude a 94.2% accounting of null deviance and examine beneficial and detrimental factors to energy-per-token costs. For agentic LLMs, we find that 93.3% of variance comes from between-session behavior for reasoning models and 89.0% for non-reasoning models. Session-length variability drives overall variance in energy usage per session. From fitting a Negative Binomial distribution to token count data, reasoning models have a variance of 35.6 million tokens squared, and non-reasoning have 5.9 million. Through a Monte Carlo simulation and Generalized Pareto Distribution, we examine tail behavior, finding that both models have approximately exponential tails. Their tail ratios are also approximately equal, indicating that they carry the same relative tail risk.

1 Introduction

In 2017, Google Brain introduced the Transformer architecture, sparking a complete overhaul at a completely separate research lab, OpenAI [8]. Researchers Alec Radford and Ilya Sutskever had been working on generative models and next-token prediction but reached the sequential bottleneck of Long Short-Term Memory (LSTM). Recursively training one word at a time was infeasible in the scope of the whole internet, a barrier which Transformers removed. Sequential chains were replaced by self-attention, with each word a constant path away from another, allowing for parallel GPU processing. Radford and Sutskever abandoned LSTM for Transformers, and in 2018, OpenAI announced GPT-1, derived from "Generative Pre-trained Transformer" [5].

As hundreds of Large Language Models have emerged into research and broader society, the notion of agentic LLMs has taken root, principally in coding. We review the key definitions and underpinnings of how modern LLMs work and the modifications that agentic LLMs make to these processes.

1.1 LLMs, Briefly Explained

At its core, training modern LLMs consists of two stages: Unsupervised pre-training and supervised fine-tuning.

Taking an unsupervised batch of tokens $\mathcal{U} = \{u_1, \ldots, u_n\}$ and context window size k, we maximize the likelihood:

$$L_1(\mathcal{U}) = \sum_i \log P(u_i | u_{i-k}, \ldots, u_{i-1}; \Theta) \tag{1}$$

A multi-layer Transformer decoder is commonly used for the language model, which first applies a self-attention operation over the input context tokens. As the name suggests, this step contextualizes a word with a window of previous tokens through its vector representation. The

vector is then put through a neural network n times, yielding h_n. Multiplying this by W_e^T and taking its softmax gives us an output distribution over target tokens.

$$h_0 = UW_e + W_p \tag{2}$$
$$h_l = \text{transformer_block}(h_{l-1}) \quad \forall l \in [1, n] \tag{3}$$
$$P(u) = \text{softmax}(h_n W_e^T) \tag{4}$$

$U = (u_{-k}, \ldots, u_{-1})$ is the context vector of tokens, n is the number of layers, W_e is the token embedding matrix, and W_p is the position embedding matrix.

Fine-tuning follows by implementing a labeled dataset

$$\mathcal{C} = \{(x^1, y^1), \ldots, (x^m, y^m)\}$$

consisting of input tokens $x^1, \ldots, x^m$ and label y. We pass the inputs through our pre-trained model to get the final transformer block's activation h_l^m. We then pass this into a new output layer with parameters W_y to predict y:

$$P(y|x^1, \ldots, x^m) = \text{softmax}(h_l^m W_y) \tag{5}$$

We then maximize the likelihood:

$$L_2(\mathcal{C}) = \sum_{(x,y)} \log P(y|x^1, \ldots, x^m) \tag{6}$$

Research has found that language modeling combined with fine-tuning as the objective improves generalization and accelerates convergence, represented by optimizing the objective with weight λ:

$$L_3(\mathcal{C}) = L_2(\mathcal{C}) + \lambda * L_1(\mathcal{C}) \tag{7}$$

Optimization of next-token prediction is the basis of these models, which has been subsequently researched and improved by increasing model size (parameters), dataset size, and compute [2].

1.2 Agentic LLMs

To define agentic LLMs, we review the underpinnings of agents and rewards in reinforcement learning [6]. We consider an agent with the goal of maximizing some objective function. They are able to perceive their environment, take an action, and receive some numerical reward as a consequence. Agents can then iterate and update optimal behavior in their environment to maximize reward over a number of steps and actions taken.

We then define agentic LLMs as large language models that have the characteristics of *reasoning*, *acting*, and *interaction* [4]. We focus our analysis on reasoning, which has been developed through the following paradigms: multi-step reasoning, self-reflection, and inference-time data retrieval.

1.3 Energy, Compute, and GPUs

A floating point operation is a basic unit of computation - one addition, subtraction, multiplication, or division - of floating point numbers. Compute is defined by the number of floating point operations (FLOP) required to find the solution [7]. For example, Google's 2023 Gemini Ultra was one of the most compute-intensive models to train ever, costing 5×10^{25} FLOP and roughly $191 million [3].

A Graphics Processing Unit (GPU) is the underpinning of how a model produces its output, and is measured in floating point operations per second, or FLOPS. The GPU will perform billions of matrix multiplications and mathematical operations, each consuming electrical power.

A joule (J) is the standard unit of energy, equating to the amount of energy consumed when a 1-watt device runs for 1 second (one joule = one watt-second).

LLMs are autoregressive, meaning that they generate one token at a time based on previous output. Interacting with the GPU hardware, the production cycle begins by loading the entire model, including its parameters, from its memory into the processing cores. The matrix multiplication and computation is relatively small, leaving GPU cores to often sit idle.

2 Methodology

We use data from ML.ENERGY, a cross-institutional initiative aimed at researching energy consumption in modern machine learning. Its benchmark v3.0 includes 46 models on NVIDIA H100 and B200 GPUs, 7 tasks, and 1,858 configurations. We scrape for LLM model data and use `energy_per_token` (J/token) as our primary efficiency metric, as it normalizes the token output of LLMs.

Table 1: LLM Dataset Characteristics

Category	**LLM Configuration**
Rows	838
Columns	35
Tasks	gpqa image-chat lm-arena-chat sourcegraph-fim video-chat
GPU Models	B200, H100
Architectures	Dense Transformer, Mamba-Hybrid, MoE
Unique Models	34

Models are classified as reasoning or non-reasoning by keyword matching on the model identifier: any model whose name contains `thinking`, `r1`, or `qwq` is designated a reasoning model, yielding 4 reasoning and 30 non-reasoning models. No rows were dropped; all 838 LLM runs are complete and stable.

For the hypothesis tests in §3.2, the unit of analysis is the per-model mean energy-per-token rather than in-

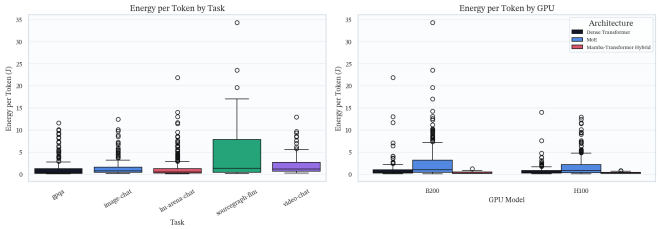


Figure 1: Distribution of Energy per Token (J) in LLMs for H100 vs. B200 architectures.

dividual runs. Each model contributes a single summary value, preventing the many repeated configurations of one model from pseudoreplicating the sample and inflating test statistics. This choice directly determines the group sizes ($n = 4$ reasoning, $n = 30$ non-reasoning) used in all subsequent tests.

3 Results

3.1 Summary Statistics

From Figure 1, per-token energy usage across tasks is relatively similar with the exception of `sourcegraph-fim` which exhibits a greater variance. Given a prefix and suffix section of code, this task asks an LLM to fill in the middle (thus, fim). Across hardware, we also see relative similarity between the 25-75% percentiles, while the newer B200 GPUs do exhibit more outliers. Across Energy per Token, Average Power, Throughput, and Latency metrics in Figure 2, we observe a significant right skew to the distribution.

3.2 Reasoning vs. Non-Reasoning Models

Non-reasoning models only undergo one forward pass through the transformer and rely on their training datasets, while reasoning models, as their name suggests, engage in what we previously described as one of the fundamental properties of agentic LLMs. Foundational developments in reasoning revolve around chain-of-thought. The methodology introduces prompts that decompose the model's workflow into intermediary steps with guardrails to prevent a spiraling into hallucination.

We test whether the differences in energy-per-token outputs from reasoning and non-reasoning models are statistically different. Table 10 notes the comprehensive statistics for every model. Given the small reasoning group ($n = 4$) and the right-skewed distribution of energy-per-token values, we apply both parametric and non-parametric tests to confirm robustness of the conclusion across distributional assumptions.

We take the mean energy-per-token results for each model and then compare their means grouped by model class. We consider a null hypothesis H_0 in which the energy-per-token outputs from reasoning and non-reasoning models are equivalent. Through Welch, Mann-Whitney, and the Permutation Test, we cannot reject our null hypothesis at an $\alpha = 0.05$ level. Thus, we proceed with the view that the energy intensity of reasoning and non-reasoning models on a normalized token output basis is not statistically different. Note that this test operates on only $n = 4$ reasoning model means, which substantially limits statistical power.

Table 2: Statistical Comparison of Energy Intensity: Reasoning vs. Non-Reasoning

Model Class	n	**Mean Energy (J/tok)**	**Std. Dev.**
Reasoning	4	3.2862	1.8692
Non-Reasoning	30	1.7760	2.4261
Test Metric		**Value**	p**-value**
Welch's t-test		$t = 1.2914$	0.2642
Mann-Whitney U		$U = 90.0$	0.1175
Permutation Test		$\Delta = 1.5102$	0.2365
Effect Size		Cohen's $d = 0.6189$	(Medium-Large)

3.3 Hardware

We design a study of tasks performed under the same conditions on both H100 and B200 GPUs. We find 150 observations with the same tasks and batch sizes across hardware, spanning 21 models and 5 unique tasks.

Table 3: Performance Delta and Statistical Significance

Metric	**Mean Δ (%)**	**Median Δ (%)**	p**-value**
Energy per Token	5.5	−1.5	0.0328*
Throughput	18.9	27.4	0.0005***
Validity Tests	**Statistic**		p**-value**
Shapiro-Wilk	$W = 0.7007$		0.0000***
Paired t-test	$t = 4.1000$		0.0001***
Wilcoxon Test	$W = 4525.0$		0.0328*

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

We find statistically significant differences in both energy-per-token and throughput from the older Hopper H100 GPU to newer Blackwell B200 hardware. We caution that the differences in energy are not normally distributed as shown in Figure 3, so we place greater emphasis on the Wilcoxon Test. The positive mean delta (+5.5%) is pulled upward by a small number of outlier configurations where B200 uses more energy. The median delta of −1.5% and the Wilcoxon result are more representative of the typical model's behavior. To a significance level of $\alpha = 0.05$, we find that B200 GPUs are statistically significantly more efficient and faster than H100s.

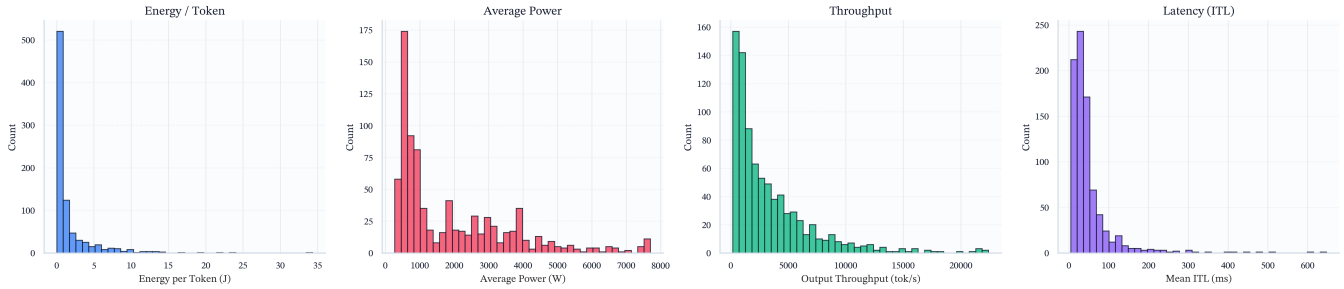


Figure 2: Distribution of key metrics: Energy (J), Power (W), Throughput (tok/s), and Latency (ms).

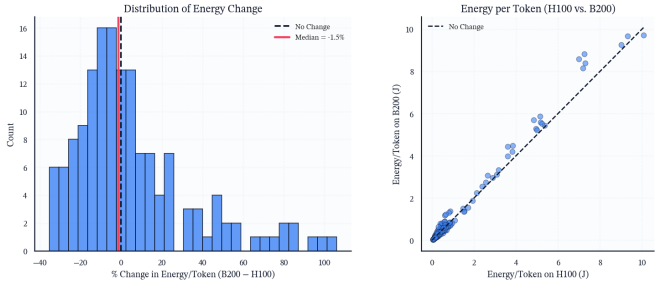


Figure 3: Change in Token-Normalized Energy Usage Across H100 and B200 GPUs.

3.4 Generalized Linear Model (GLM)

We model `energy_per_token_joules` as a function of software and hardware factors to fit a Gamma GLM with log link. GLM extends the linear OLS so that it can have a response variable that follows non-normal distributions. A Gamma regression is appropriate because energy violates two of the assumptions in OLS:

- Energy is never a negative value. While OLS can predict negative values, a Gamma regression with log link is always greater than 0.
- The histogram of `energy_per_token` is right-skewed, fitting a Gamma regression better than OLS, which assumes symmetric Normal errors.

We examine the *residual deviance* of this GLM, which serves as a measure of goodness-of-fit.

A *saturated model* has an equal number of observations and parameters. Thus, for each observation i, $\hat{\mu} = y_i$. In this perfect fit, our log-likelihood $\ell(\mathbf{y}; \boldsymbol{\mu}) = \ell(\mathbf{y}; \mathbf{y})$. A *fitted model* has fewer parameters than number of observations. Using a link function, we constrain $\boldsymbol{\mu}$ to be a function of the predictors: $g(\mu) = \mathbf{X}\beta$. So the log-likelihood is denoted as $\ell(\mathbf{y}; \hat{\boldsymbol{\mu}})$.

Then our *residual deviance* measures the distance between the fitted model and the saturated model.

$$D_{res}(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2\left[\ell(\mathbf{y}; \mathbf{y}) - \ell(\mathbf{y}; \hat{\boldsymbol{\mu}})\right].$$

We also define the *null deviance* as the deviance of the null model, which only contains an intercept and no predictors. Thus, $\hat{\mu}_i = \bar{y}$, and we have

$$D_{null} = 2\left[\ell(\mathbf{y}; \mathbf{y}) - \ell(\mathbf{y}; \bar{\mathbf{y}})\right].$$

We calculate this as such: for an observation $y_i > 0$, the PDF is:

$$f(y_i; \mu_i, \nu) = \frac{1}{\Gamma(\nu)} \left(\frac{\nu y_i}{\mu_i}\right)^{\nu} \frac{1}{y_i} \exp\left(-\frac{\nu y_i}{\mu_i}\right)$$

where ν is the shape parameter. Then the log-likelihood function is

$$\ell(\mu_i, \nu; y_i) = \nu \ln(\nu) + \nu \ln(y_i) - \nu \ln(\mu_i) - \ln \Gamma(\nu) - \ln(y_i) - \frac{\nu y_i}{\mu_i}$$

Note that terms $\nu \ln(\nu)$, $\ln \Gamma(\nu)$, and $\nu \ln(y_i) - \ln(y_i)$ are constants relative to the mean. Assuming ν is constant, the saturated model is

$$\ell(y_i) = \text{const} - \nu \ln(y_i) - \frac{\nu y_i}{y_i} = \text{const} - \nu \ln(y_i) - \nu$$

and the fitted model is

$$\ell(\hat{\mu}_i) = \text{const} - \nu \ln(\hat{\mu}_i) - \frac{\nu y_i}{\hat{\mu}_i}.$$

Thus,

$$\ell(y_i) - \ell(\hat{\mu}_i) = (-\nu \ln(y_i) - \nu) - \left(-\nu \ln(\hat{\mu}_i) - \frac{\nu y_i}{\hat{\mu}_i}\right)$$

$$\ell(y_i) - \ell(\hat{\mu}_i) = -\nu \ln(y_i) + \nu \ln(\hat{\mu}_i) + \frac{\nu y_i}{\hat{\mu}_i} - \nu$$

$$\ell(y_i) - \ell(\hat{\mu}_i) = \nu \left[-\ln\left(\frac{y_i}{\hat{\mu}_i}\right) + \frac{y_i}{\hat{\mu}_i} - 1\right]$$

Since ν appears as a common factor in both D_{res} and D_{null}, it cancels when computing the deviance explained ratio $1 - D_{\text{res}}/D_{\text{null}}$. Defining the unit deviance by factoring out ν, our deviance is:

$$D = 2\sum_{i=1}^{n} \left[-\ln\left(\frac{y_i}{\hat{\mu}_i}\right) + \frac{y_i - \hat{\mu}_i}{\hat{\mu}_i}\right].$$

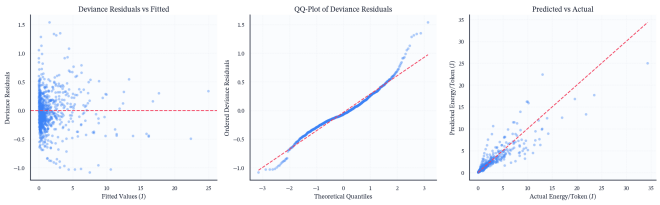


Figure 4: Distribution of Residual Deviance from `energy_per_token` data.

Table 4: GLM Goodness of Fit

Metric	Value	
Null deviance	1573.44	
Residual deviance	90.49	
Deviance explained		94.2%

Given that the model accounts for 94.2% of the null deviance, we conclude that the GLM is a good fit of per-token energy behavior. That is, a small set of factors highly determine the energy-per-token behavior of LLMs.

Figure 5 shows beneficial and detrimental model construction decisions that factor into energy behavior. A 1-unit increase in the number of log-parameters multiplies energy-per-token by 1.94x. Using `mxfp4` - 4-bit microscaling floating point - which uses significantly less energy than the baseline `bfloat16` 16-bit, improves energy usage by 60%. Mixture of Experts (MoE) models only incorporate a subset of parameters for each token, which is revealed in their 33% improvement per token over Dense Transformer models.

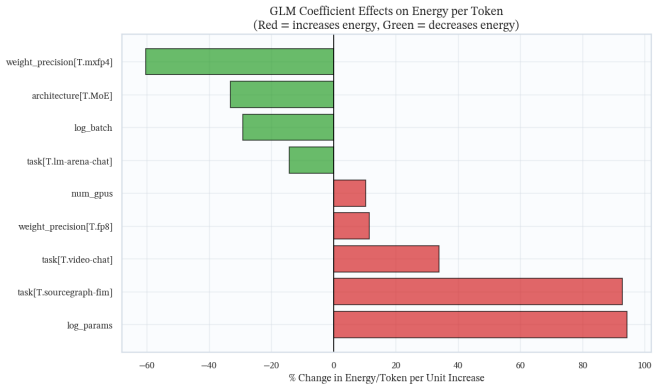


Figure 5: Effects of the GLM coefficients on Energy-per-Token Outputs

Table 5: Gamma GLM Coefficients: Predictors of Energy per Token (J)

Predictor	Coef (β)	Exp(β)	% Effect	p-value
Model Features				
Log(Parameters)	0.6648	1.9442	94.42	0.0000***
Num GPUs	0.0991	1.1042	10.42	0.0000***
Architecture: MoE	−0.4016	0.6693	−33.07	0.0000***
Tasks (Ref: Image-Chat)				
Code Completion (FIM)	0.6564	1.9278	92.78	0.0000***
Video Chat	0.2923	1.3395	33.95	0.0000***
Arena Chat	−0.1539	0.8573	−14.27	0.0001***
Precision & Optimization				
Log(Batch Size)	−0.3449	0.7083	−29.17	0.0000***
Precision: FP8	0.1091	1.1152	11.52	0.0078**
Precision: MXFP4	−0.9246	0.3967	−60.33	0.0000***

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

3.5 Compound Distribution Models for Agentic LLMs

In an agentic session, the agent issues N model calls, each generating L_i output tokens, which consume energy e_t per token.

$$E_{\text{session}} = \sum_{i=1}^{N} L_i \cdot e_t \tag{8}$$

This presents an interesting examination of tail behavior from agentic models, since the total task energy is a compound random variable. We present a two-level stochastic model, consisting of:

1. Gamma distribution-fitted per-token energy, representing the inner component
2. Negative Binomial distribution-fitted output token counts per request, representing the outer component

Note that session lengths N are estimations of how many requests may run in a session as a fixed distribution across both reasoning and non-reasoning models because the dataset only records individual runs and not full agentic sessions. Analysis of the convolution can then be estimated by Monte Carlo simulation to characterize the compound tails. We refine our Gamma distributions and focus on whether token-count variability affects tail risk through the lens of energy cost.

Applying Gamma distributions to all reasoning and non-reasoning models in our dataset, we find that their profiles for `energy_per_token_joules` are statistically different. This is not a contradiction of §3.2: the LRT here operates on 60 vs. 778 individual runs and detects a difference in distributional shape, whereas the §3.2 group-mean tests had only $n = 4$ reasoning model means and lacked the power to reject H_0. This applies to equation 8, as each simulated token-count L_i is multiplied by $\bar{e}_t = \frac{\alpha}{\beta}$ to estimate per-request energy E_i.

To capture the reasoning attributes of agentic LLMs, we examine individual results for an LLM task known as `GPQA`: Graduate-Level Google-Proof Q&A. The benchmark contains hundreds of specialized STEM questions designed to require deep reasoning even with internet context. The goal is to interpret some result from comparing how reasoning and non-reasoning models behave when presented with this group of tasks. We capture 60 runs of reasoning models spanning 16,434 requests and 127 runs

Table 6: Parameters for Session Energy Modeling

Variable	Distribution
Session Length (N)	$N \sim \text{NB}(\mu = 10, r = 2)$
Tokens per Call (L_i)	$L_i \sim \text{NB}(\hat{\mu}, \hat{r})$
Energy per Token (e_t)	$e_t \sim \text{Gamma}(\hat{\alpha}, \hat{\beta})$

Table 7: Gamma Distribution Parameter Estimates: Reasoning vs. Non-Reasoning

Group	n	Shape (α)	Rate (β)
Reasoning	60	0.9694	0.3304
Non-Reasoning	778	0.6448	0.4112
Statistical Test	**Value**		***p*-value**
LRT	$\chi^2 = 22.697$		$< 0.0001^{***}$

*** $p < 0.001$. Reject H_0: Groups exhibit distinct shape characteristics.

of non-reasoning models spanning 69,300 requests.

Table 8: Token Count Distribution

Group	n	Mean	Std Dev	Var/Mean	NB (r)
Reasoning†	16,434	9085.1	5946.9	3892.73	2.32
Non-Reasoning	69,300	2821.5	3189.9	3606.44	1.35
Likelihood Ratio Test (vs. Poisson)					
Reasoning	$\chi^2 = 6.07 \times 10^7$			$p < 0.001^{***}$	
Non-Reasoning	$\chi^2 = 1.66 \times 10^8$			$p < 0.001^{***}$	

† Note: 0.32% of Reasoning requests hit the 32,768 token context ceiling.

Overdispersion refers to the variance of a count variable being larger than its mean. For reference, a Poisson distribution assumes that the mean equals variance. Contrastingly, the Negative Binomial distribution introduces a dispersion parameter r to model the relationship between the mean and variance:

$$\text{Var}[L] = \mu + \frac{\mu^2}{r} \tag{9}$$

As $r \to \infty$, the dispersion term $\frac{\mu^2}{r} \to 0$, causing the distribution to collapse into a Poisson distribution where $\text{Var}[L] = \mu$. Conversely, as r decreases, the variance greatly exceeds the mean, capturing overdispersion. From Table 8, we observe overwhelming evidence to favor a Negative Binomial distribution, as both reasoning and non-reasoning models exhibit a Variance/Mean of 3892 and 3606, respectively. While the parameter r for reasoning models is greater than that of non-reasoning models, we calculate their respective variances to be 35.6 million tokens squared and 5.9 million tokens squared. Figure 6 demonstrates our cumulative probability functions, showing that reasoning models are more likely to produce more output tokens per request.

Running 50,000 simulated sessions, N is drawn from its Negative Binomial distribution, and then we draw N token count samples from our Negative Binomial distribution of L, each multiplied by $\bar{e}_t$, and then all summed.

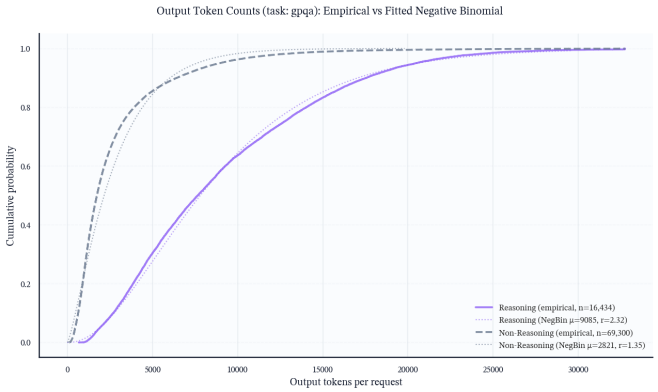


Figure 6: Empirical and Fitted Negative Binomial Distributions of Reasoning and Non-Reasoning Models for `GPQA`

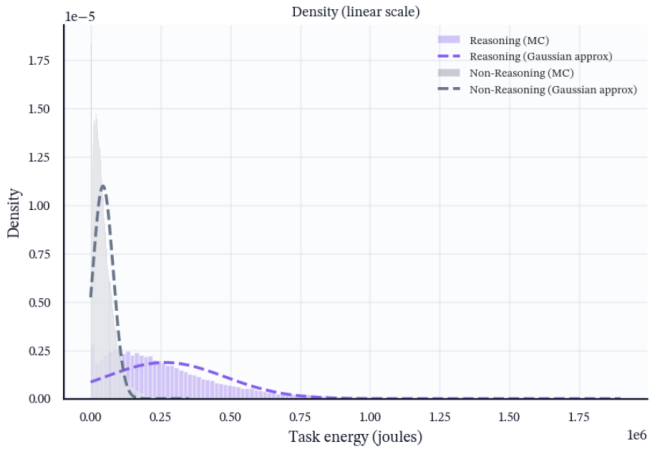


Figure 7: Compound Energy Distribution with Monte Carlo Simulation

The Law of Total Variance is used to examine the breakdown of uncertainty across agentic LLMs. If X and Y are random variables on the same probability space, and Y has finite variance, then:

$$\text{Var}(Y) = E[\text{Var}(Y|X)] + \text{Var}(E[Y|X])$$

Conditioning on N, we solve for $E[E_{\text{session}}|N]$. If $N = n$ is known, the session consists of n i.i.d. requests. By linearity of expectation:

$$E[E_{\text{session}}|N = n] = n \cdot E[E_i] = n \cdot \mu_L \cdot \bar{e}_t$$

So, generalizing for random variable N:

$$E[E_{\text{session}}|N] = N \cdot E[E_i]$$

We then solve for $\text{Var}[E_{\text{session}}|N]$. Given $N = n$ and assuming the E_i are independent:

$$\text{Var}[E_{\text{session}}|N = n] = n \cdot \text{Var}[E_i]$$

$\text{Var}[E_i]$ comes from the Negative Binomial variance of the token counts L_i:

$$\text{Var}[E_i] = \bar{e}_t^2 \cdot \text{Var}[L_i] = \bar{e}_t^2 \left(\mu_L + \frac{\mu_L^2}{r_L} \right)$$

So as a function of N,

$$\text{Var}[E_{\text{session}}|N] = N \cdot \text{Var}[E_i]$$

Finally, plugging back into the definition:

$$\text{Var}[E_{\text{session}}] = E[N \cdot \text{Var}[E_i]] + \text{Var}[N \cdot E[E_i]]$$

Since $\text{Var}[E_i]$ and $E[E_i]$ are constants determined by fitted model parameters and are not functions of N, we have

$$\text{Var}[E_{\text{session}}] = \underbrace{E[N] \cdot \text{Var}[E_i]}_{\text{Within-session}} + \underbrace{\text{Var}[N] \cdot (E[E_i])^2}_{\text{Between-session}} \quad (10)$$

$E[N] \cdot \text{Var}[E_i]$ represents the within-session uncertainty. $E[N]$ is the average number of requests per session, and $\text{Var}[E_i]$ is the variance in the energy consumed by a single request. $\text{Var}[N] \cdot (E[E_i])^2$ represents the between-session uncertainty, hinging on the variance of how long an agentic loop lasts. Our results indicate that 93.3% of variance comes from between-session for reasoning models and 89.0% for non-reasoning models. Thus, session-length variability is the primary driver of overall variance in energy usage per task. Figure 7 displays an increased dispersion in reasoning models compared to non-reasoning, as well as right skew in both compound distributions as a result of occasional long requests or outputs. Note that due to the larger task energy usage data in reasoning models, the x-axis scale causes the non-reasoning model distribution to appear more compressed.

Figure 8, which is in log scale, and fitting a Generalized Pareto Distribution (GPD) allow us to extrapolate tail behavior beyond the P99 region. The GPD is fit via Peaks Over Threshold (POT) to the top 10% of simulated sessions (5,000 exceedances above the P90 threshold), and then extrapolated to P99 and beyond where only approximately 500 of our 50,000 Monte Carlo samples lie. The shape parameter ξ of the GPD determines the heaviness of the distribution's tail:

- $\xi > 0.1 \rightarrow$ **Heavy Pareto tail**: Extreme events can become arbitrarily severe, indicating significant energy risk.
- $\xi \approx 0 \rightarrow$ **Exponential tail**: Represents moderate risk, where the probability of extreme events decays exponentially.
- $\xi < -0.1 \rightarrow$ **Bounded thin tail**: Indicates a hard physical or logical ceiling on how high the energy consumption can go.

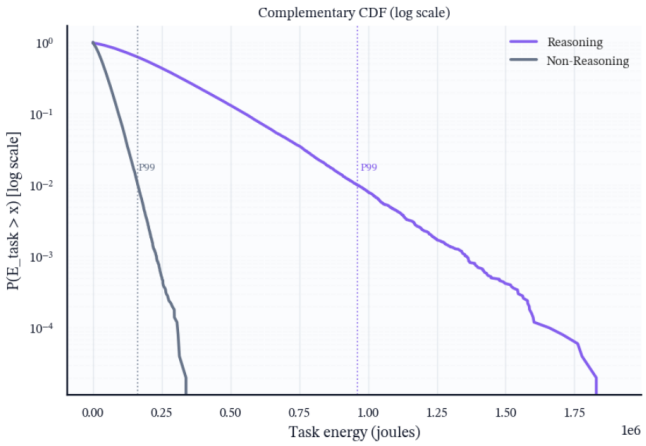


Figure 8: Complementary CDF of Reasoning and Non-Reasoning Distributions in Log Scale

Fitting the GPD yields $\xi = -0.043$ $[-0.069, -0.019]$ for reasoning and $\xi = -0.036$ $[-0.062, -0.012]$ for non-reasoning. These are approximately exponential, and crucially not indicative of heavy tail risk behavior. Figure 8 bolsters this result with the approximately vertical tail ends of both complementary CDFs. A straight diagonal line in log space indicates an exponential tail, while a horizontally flattening slope would represent heavy Pareto tails.

We define Tail Ratio as

$$\frac{\text{energy at that percentile}}{\text{median session energy}}$$

Practically, we use this ratio to consider worst-case scenarios if planning capacity based on the median session. For example, if we planned a GPU cluster for a reasoning model based on the median session power draw, in one in every 100 sessions, we would go 4.43x over budget. Interestingly, we find that tail ratios are almost exactly similar for reasoning and non-reasoning models. That is, they carry the same relative tail risk.

Table 9: Tail Risk Estimates

Group	**%**	**Energy (J)**	**Tail Ratio**
Non-Reasoning	P75	6.19×10^4	1.74
	P95	1.15×10^5	3.23
	P99	1.63×10^5	4.58
	P99.9	2.31×10^5	6.47
Reasoning	P75	3.69×10^5	1.70
	P95	6.82×10^5	3.14
	P99	9.61×10^5	4.43
	P99.9	1.36×10^6	6.26

4 Conclusion

A comprehensive study of pricing in LLMs over the past six years underlines the bifurcation our results show between non-reasoning and reasoning models [1]. We distinguish between different models based on their intelligence tier. Economy and mid-tier models are precipitously declining in pricing, with half-lives of 1.10 and 1.55 years, respectively. Meanwhile, flagship tier models show virtually no exponential price decay, with an $R^2 = 0.031$. In fact, flagship models command a reasoning premium averaging 31.5 times non-reasoning prices. Research also supports our results on software's effect on energy costs. Total factor productivity residuals, which capture software and architectural innovation, account for 103.7% of cost reductions. Conversely, GPU hardware contributed -0.9% to cost decomposition. Other explorations also indicate a training-inference elasticity of $\beta = 0.432$. So, for a 1% increase in training costs comes an inference pricing increase of 0.43%. The econometric analysis above was based on a $ per million input tokens metric, but its implications are analogous to our work with output tokens and energy costs.

Non-reasoning models are becoming commoditized, with pricing now dictated by competition rather than technological advancements. Innovations in pre-training are less computationally intensive than the implications of recurring request loops and token usage in agentic sessions. Nevertheless, our results imply promising indications for planning for the future behavior of the session-level energy usage distribution. An approximately exponential tail that displays a vertical complementary CDF means that there is low risk of extreme energy usage tasks. While token usage and agentic loops are likely to increase in the future, current data provides solace in that explosive energy usage from runaway sessions is not, at an individual level, an extreme risk.

Promising directions for future research include a macro analysis of how exogenous events can affect coordinated spikes in agentic usage. For example, thousands of financial services agentic LLMs could be rapidly forced into right tail behavior if unforeseen news or a data print came out. Alternatively, interesting approaches could be beneficial to understanding regional quirks to agentic usage. Future work is to be done in analyzing regional behavior in San Francisco, for example, where there is arguably the highest density of coding agent usage in the world. In May 2026, GitHub announced that they would be shifting their popular Copilot service that aggregates various coding agents from a monthly subscription with no token-counting and a density-based premium request limit to a token-based pricing plan. This, and advances in tradable benchmarks for GPU compute such as Ornn and Silicon Data, imply a fungible future for high-density token usage.

Other directions for future work include examining session lengths N and token counts L_i as a joint distribution. In our work, since our data only included isolated request-level outputs, we were constrained and had to estimate a fixed distribution for N. In reality though, N and L_i can be highly endogenous, as a reasoning model could efficiently use little tokens and few sessions, or conversely, spiral into multiple sessions with drawn out token lengths.

Lastly, we also consider the next application domain of agentic artificial intelligence. In the biological sciences, applying AI has become a hotbed of recent advancement and introduces a physical resource constraint to the wet lab.

References

[1] Mingdeng Du. Tiered super-Moore's law: Price evolution, production frontiers, and market competition in large language model inference services. *arXiv preprint arXiv:2603.28576*, 2026.

[2] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

[3] Iyngkarran Kumar and Sam Manning. Trends in frontier AI model count: A forecast to 2028. *arXiv preprint arXiv:2504.16138*, 2025.

[4] Aske Plaat, Max van Duijn, Niki van Stein, Mike Preuss, Peter van der Putten, and Kees Joost Batenburg. Agentic large language models, a survey. *arXiv preprint arXiv:2401.03428*, 2024.

[5] Alec Radford, Karthik Narasimhan, Tim Salimans, and Ilya Sutskever. Improving language understanding by generative pre-training. Technical report, OpenAI, 2018.

[6] Stuart J. Russell and Peter Norvig. *Artificial Intelligence: A Modern Approach*. Pearson, 4th edition, 2020.

[7] Ryan Tibshirani. Convex optimization, Lecture 19. Lecture Notes, Carnegie Mellon University, November 2015. Scribes: Bohan Li, Donghan Yu, and Ge Huang.

[8] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, pages 5998–6008, 2017.

A Summary of Model Energy Statistics

Table 10: Energy Statistics for LLM Inference (ML.ENERGY v3.0)

Model	**Type**	n_{obs}	**Mean (J)**	**Std Dev**	**Skew**	**Ex. Kurtosis**
Non-Reasoning Models						
Qwen 3 14B	Non-Reasoning	26	0.3465	0.3088	1.6049	2.3599
Qwen 3 VL 30B A3B Instruct	Non-Reasoning	29	0.5138	0.2604	0.0210	−1.2646
Qwen 3 VL 235B A22B FP8	Non-Reasoning	62	3.2575	3.1814	1.3964	1.0675
Qwen 3 VL 235B A22B	Non-Reasoning	57	2.6357	2.2007	1.1500	0.4154
Qwen 3 Omni 30B A3B	Non-Reasoning	27	0.7166	0.3942	0.1711	−1.0644
Qwen 3 Coder 480B A35B FP8	Non-Reasoning	16	12.1947	7.9288	1.3469	1.3530
Qwen 3 Coder 480B A35B	Non-Reasoning	10	4.9228	3.9421	0.8834	−0.5875
Qwen 3 Coder 30B A3B	Non-Reasoning	24	0.4170	0.2514	0.8945	−0.2114
Qwen 3 8B	Non-Reasoning	29	0.2535	0.2415	2.1148	4.9459
Qwen 3 32B	Non-Reasoning	31	0.7042	0.5927	1.2201	0.5181
Qwen 3 30B A3B Instruct	Non-Reasoning	31	0.2611	0.2097	1.2248	0.3782
Qwen 3 235B A22B FP8	Non-Reasoning	23	3.0981	3.0207	1.4397	0.9461
Qwen 3 235B A22B	Non-Reasoning	31	2.2511	2.2717	1.2065	0.3645
Qwen 3 VL 32B Instruct	Non-Reasoning	24	0.7972	0.5028	1.0531	−0.0595
NV Nemotron Nano 9B V2	Non-Reasoning	29	0.2602	0.1687	1.6640	2.2698
Qwen 3 VL 8B Instruct	Non-Reasoning	24	0.3055	0.1545	0.6291	−0.7359
NV Nemotron Nano 12B V2	Non-Reasoning	27	0.3482	0.2579	1.7562	2.8333
DeepSeek V3.1	Non-Reasoning	17	3.4454	2.9396	0.8933	−0.4790
GPT OSS 120B	Non-Reasoning	39	0.2346	0.1952	1.1633	0.4145
GPT OSS 20B	Non-Reasoning	26	0.0952	0.0824	1.8568	2.6149
Gemma 3 12B	Non-Reasoning	26	0.3154	0.1989	1.0933	0.0375
Gemma 3 27B	Non-Reasoning	32	0.6765	0.4478	0.9516	−0.4282
Llama 3.1 405B Instruct	Non-Reasoning	12	4.6808	5.9792	2.0032	2.8971
Llama 3.1 405B FP8	Non-Reasoning	15	4.6528	3.9236	1.3926	0.6469
Llama 3.1 70B Instruct	Non-Reasoning	22	0.9406	0.9916	1.8808	2.4111
Llama 3.1 8B Instruct	Non-Reasoning	20	0.1400	0.1102	1.7170	1.8790
Llama 3.3 70B Instruct	Non-Reasoning	19	1.0664	1.0431	1.7072	1.7051
Llama 4 Maverick 17B FP8	Non-Reasoning	15	1.3308	0.7559	1.0699	0.0795
Llama 4 Scout 17B 16E	Non-Reasoning	32	1.7707	1.4396	1.1937	0.4416
NV Nemotron Nano 12B VL	Non-Reasoning	3	0.6483	0.1392	0.2946	−1.5000
Reasoning Models						
Qwen 3 30B Thinking	Reasoning	15	0.4897	0.3263	0.9431	0.3382
Qwen 3 235B Thinking FP8	Reasoning	22	3.7934	3.2003	0.9431	−0.2134
Qwen 3 235B Thinking	Reasoning	18	3.1507	2.3236	0.6663	−0.7358
DeepSeek R1	Reasoning	5	5.7111	2.6085	0.3542	−1.1004